Sahaj Bhadja

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Summary

Infrastructure-focused software engineer with 2+ years supporting cloud services, networked applications, automation, and production operations. Experienced in AWS, Azure, Linux, Python, Bash, TCP/IP, Docker, Kubernetes, Terraform, Ansible, CI/CD, monitoring, and incident troubleshooting.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built Python services on Azure SQL and Blob Storage for 500+ users, tuning cloud resources, indexed queries, and secure access controls to reduce infrastructure-backed latency from 1.2 seconds to 350 milliseconds.
- Operated 5,000+ monthly background jobs using explicit states, retries, reconciliation, and failure isolation, tripling throughput while improving service monitoring, incident diagnosis, recovery workflows, and production reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Deployed Flask services and networked application workflows handling 10,000+ daily requests, implementing secure file ingestion, health checks, logging, and AWS RDS connectivity for 150+ production users.
- Created concurrent processing for 1,000+ files with automated validation, AWS RDS persistence, and structured failure handling, reducing turnaround 60% while maintaining stable infrastructure behavior during peak workloads.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built reproducible Linux-based simulation and data pipelines processing 100,000+ records, adding automated validation, runtime instrumentation, structured logs, and systematic troubleshooting to improve planning accuracy 18% across controlled evaluations.
- Documented system architecture, experiment configurations, and operational procedures while tuning distributed processing workflows that improved closed-loop stability 15% across repeated simulations and accelerated issue reproduction for researchers.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed cloud integrations and ETL automation using Python, AWS Glue, S3, SQL, and REST services, improving response times 30% and infrastructure-backed processing efficiency 68% across production workflows.
- Maintained 20+ automated test suites with Jenkins, PyTest, and Mockito across Windows, Linux, and macOS, accelerating incident troubleshooting, release validation, and overall confidence in cross-platform production deployments.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular Node.js services and WebSocket workflows across eight authentication, permissions, notifications, and operational modules, synchronizing 2 million+ Azure SQL records through scalable, independently maintainable service boundaries.
- Automated Docker builds, validation, staged deployments, and production monitoring through Jenkins and Azure, reducing release time 50% while improving configuration consistency, service availability, and overall operational reliability.

Skills

Cloud & Infrastructure: AWS, Azure, EC2, ECS/Fargate, AWS RDS, AWS Glue, Amazon S3, Blob Storage, virtual machines, cloud services, infrastructure operations
Networking & Systems: Linux, Bash, TCP/IP, network protocols, client-server systems, distributed systems, REST APIs, WebSockets, networked services, system troubleshooting
Automation & Containers: Python, Terraform, Ansible, Docker, Kubernetes, Jenkins, Git, CI/CD, infrastructure as code, scripting, deployment automation, configuration management
Operations & Data: PostgreSQL, MySQL, Azure SQL, MongoDB, monitoring, logging, observability, incident response, root-cause analysis, documentation, PyTest, unit testing, integration testing

Projects

FinNexus | Live | *TypeScript, REST APIs, WebSockets, MongoDB, Docker, Kubernetes*
- Deployed a containerized platform supporting 50,000+ users and one million transactions, integrating REST services, WebSocket networking, MongoDB, Kubernetes, monitoring, and automated CI/CD workflows for scalable production operations.

Trend Tracker | Live | *AWS EC2, React, Python, CNN, Strapi*
- Deployed a cloud application on AWS EC2 serving 3 million+ users, integrating Python inference services, React workflows, persistent data services, operational monitoring, and scalable Linux-based production infrastructure.

Chat2SQL | Code | *Java, TypeScript, PostgreSQL, Docker, Maven, Linux*
- Built Linux-compatible containerized workflows supporting 10+ databases, combining Docker, Maven, PostgreSQL, Java services, configuration validation, local debugging, and repeatable automated deployment processes across Windows, macOS, and Linux.